16 ACIDS, ALKALIS AND TITRATIONS

This chapter explores what indicators, acids and alkalis are, and how to carry out an acid–alkali titration. The reactions of acids are also discussed in Chapter 17.



▲ Figure 16.1 Acids range from the extremely dangerous, needing protective clothing to clean up spills . . .



▲ Figure 16.2 . . . to a natural part of our diet: oranges contain citric acid.

LEARNING OBJECTIVES

- Describe the use of litmus, phenolphthalein and methyl orange to distinguish between acidic and alkaline solutions
- Understand how the pH scale, from 0–14, can be used to classify solutions as strongly acidic (0–3), weakly acidic (4–6), neutral (7), weakly alkaline (8–10) and strongly alkaline (11–14)
- Describe the use of universal indicator to measure the approximate pH value of an aqueous solution
- Know that acids in aqueous solution are a source of hydrogen ions, and alkalis in an aqueous solution are a source of hydroxide ions
- Know that alkalis can neutralise acids

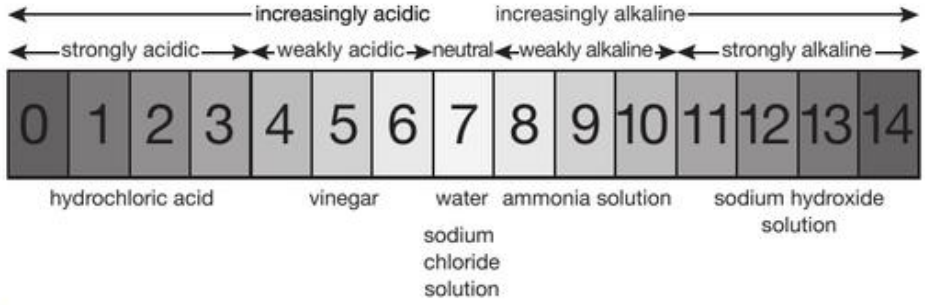
CHEMISTRY ONLY

- Describe how to carry out an acid–alkali titration.

pH AND INDICATORS

THE pH SCALE

The pH scale ranges from about 0 to about 14, and tells you how acidic or how alkaline a solution is.



▲ Figure 16.3 The pH scale. Vinegar is a solution of ethanoic acid.

We can classify substances as strongly acidic/alkaline or weakly acidic/alkaline based on their pH. Remember, when writing pH, it is a small p and a big H (the symbol for hydrogen).

Table 16.1 We can classify solutions according to their pH

	pH	Solution
strongly acidic	0–3	hydrochloric acid
weakly acidic	4–6	ethanoic acid (vinegar)
neutral	7	sodium chloride
weakly alkaline	8–10	ammonia
strongly alkaline	11–14	sodium hydroxide

EXTENSION WORK

These are actually only approximate ranges and depend on the concentration of the acid/alkali solution. A 0.10 mol/dm^3 solution of ethanoic acid actually has a pH of 2.88.

MEASURING pH

USING UNIVERSAL INDICATOR

HINT

You can measure pH much more accurately using a pH meter.

Universal indicator is made from a mixture of indicators, which change colour in a gradual way over a range of pH values. It can be used as a solution or as paper. The most common form is known as *full-range* universal indicator. It changes through a variety of colours from pH 1 right up to pH 14, but it isn't very accurate.



▲ Figure 16.4 Using universal indicator solution to measure the pH of various solutions.

The colour of the paper or solution is checked against a chart to find the pH.

ACID–ALKALI INDICATORS

Any substance that has different colours depending on the pH can be used as an indicator. Some common indicators and their colours in acidic and alkaline solutions are shown in Table 16.2.

Table 16.2 The colours of various indicators in acidic and alkaline solutions

	Acid	Alkali
litmus	red	blue
methyl orange	red	yellow
phenolphthalein	colourless	pink
universal indicator	red	blue

KEY POINT

Litmus can be used as a solution or as litmus paper.

Litmus is red in acidic solutions and blue in alkaline ones. In neutral solutions, the colour is purple, which is an equal mixture of the red and blue forms. Universal indicator is green in neutral solutions.

ACIDS

Table 16.3 Some acids showing the replaceable hydrogen

Acid	Formula
hydrochloric acid	HCl
nitric acid	HNO_3
sulfuric acid	H_2SO_4
ethanoic acid	CH_3COOH
phosphoric acid	H_3PO_4

The formulae of some acids are given in Table 16.3.

All acids contain hydrogen and when acids react the hydrogen shown in red in Table 16.3 is replaced by something else; all acids have replaceable H. For example, when hydrochloric acid reacts with sodium hydroxide we obtain:

$$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$$

The H of the HCl has been replaced by an Na.

Not all of the hydrogens in acids are replaceable, for example in ethanoic acid only the H attached to the O is replaceable, not the ones joined to the C. The reactions of carboxylic acids such as ethanoic acid are discussed further in Chapter 27.

When acids are in water they dissociate (break apart) to form hydrogen ions (H^+), for example:

$$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$$
$$HNO_3(aq) \rightarrow H^+(aq) + NO_3^-(aq)$$
$$H_2SO_4(aq) \rightarrow 2H^+(aq) + SO_4^{2-}(aq)$$

EXTENSION WORK

At the next level, you will discover that pH is defined by a mathematical equation: $pH = -\log_{10}[H^+(aq)]$.

When we are measuring pH we are actually measuring the concentration of these H^+ ions in the solution. This is why the H in pH is written with a capital letter.

We can define acids as *substances that act as a source of hydrogen ions (H^+) in solution*.

BASES

Bases are substances that neutralise acids by combining with the hydrogen ions in them. When we are referring to a base at this level we usually mean a metal oxide, a metal hydroxide or ammonia (NH_3).

ALKALIS



▲ Figure 16.5 Ammonia solution is sometimes called ammonium hydroxide. It is an alkali and is used in some cleaning products. Some universal indicator solution has been added to it in the beaker.

Some bases dissolve in water to form solutions containing hydroxide ions. These are alkalis.

Alkalis are a source of hydroxide (OH^-) ions in solution.

Examples of alkalis are sodium hydroxide and potassium hydroxide (all the Group 1 hydroxides). When sodium hydroxide is in water it breaks apart to form sodium and hydroxide ions:

$$NaOH(aq) \rightarrow Na^+(aq) + OH^-(aq)$$

The other alkali you will meet is a solution of ammonia (NH_3). The ammonia reacts with the water to form ammonium ions and hydroxide ions:

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

These alkalis all have a pH greater than 7.

OTHER ALKALINE SOLUTIONS

There are some other substances, such as soluble metal carbonates, that react with water to form hydroxide ions:

$$Na_2CO_3(aq) + H_2O(l) \rightleftharpoons NaOH(aq) + NaHCO_3(aq)$$